

PY32F030系列应用指南

-中断向量重新映射



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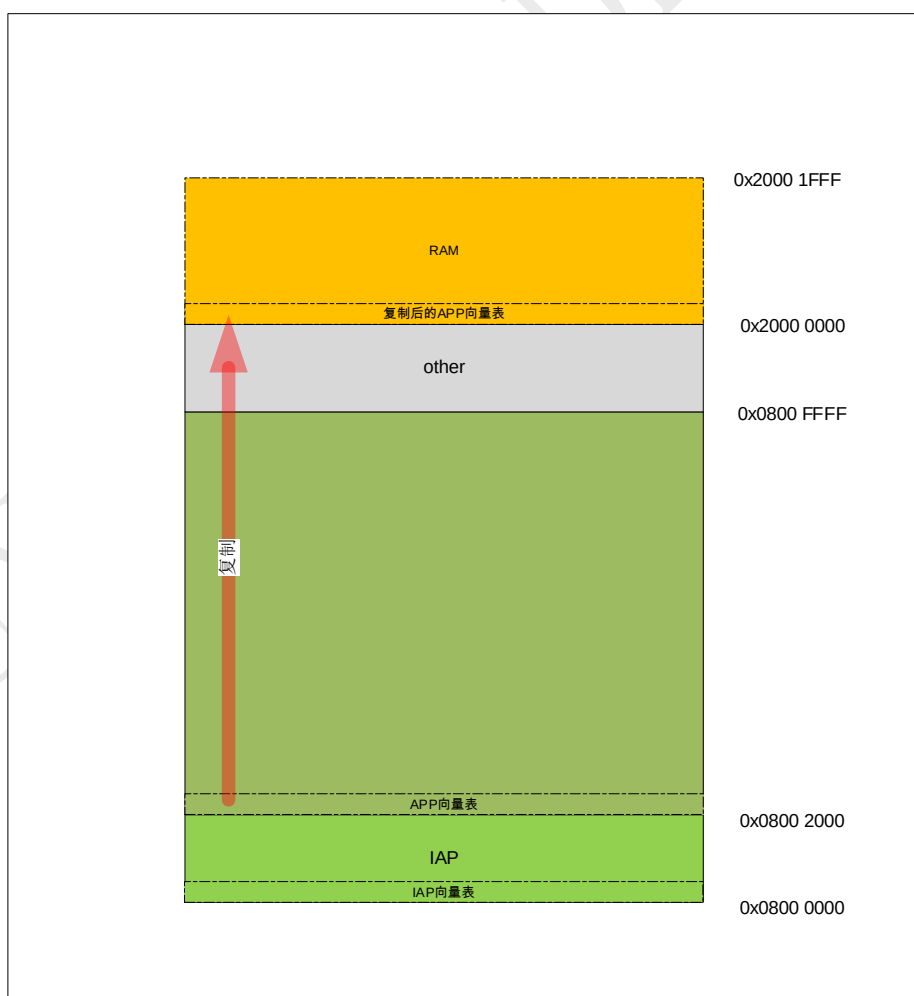
1 简介

PY32F030 系列微控制器采用高性能的 32 位 ARM®Cortex®-M0+内核，在系统复位后的第 4 个系统时钟进行锁存启动模式，常规使用的启动模式是从Main flash启动，即 0x800 0000 跟启动存储空间的0x0000 0000 对齐。从地址 0x0000 0000 取堆栈顶的值，然后从启动存储器的 0x0000 0004 地址开始执行指令。

对于部分应用场景，客户的代码并不都在Main flash的起始地址(0x800 0000)运行。例如用户自定义的引导程序（IAP）实现用户应用程序（APP）的升级，就需要用到中断向量表的重新映射。此应用笔记的目的是讲述如何实现中断向量表的重新映射。

1.1 实现原理

PY32F030系列需要将中断向量表从内部FLASH复制到RAM后，再设置SCB->VTOR寄存器，这样操作后 APP才能正常运行。示意图如下，将APP向量表复制到RAM的起始地址，并将SCB->VTOR寄存器同样设置为RAM的起始地址



具体可参考demo例程，IAP和APP代码。

1.2 IAP代码

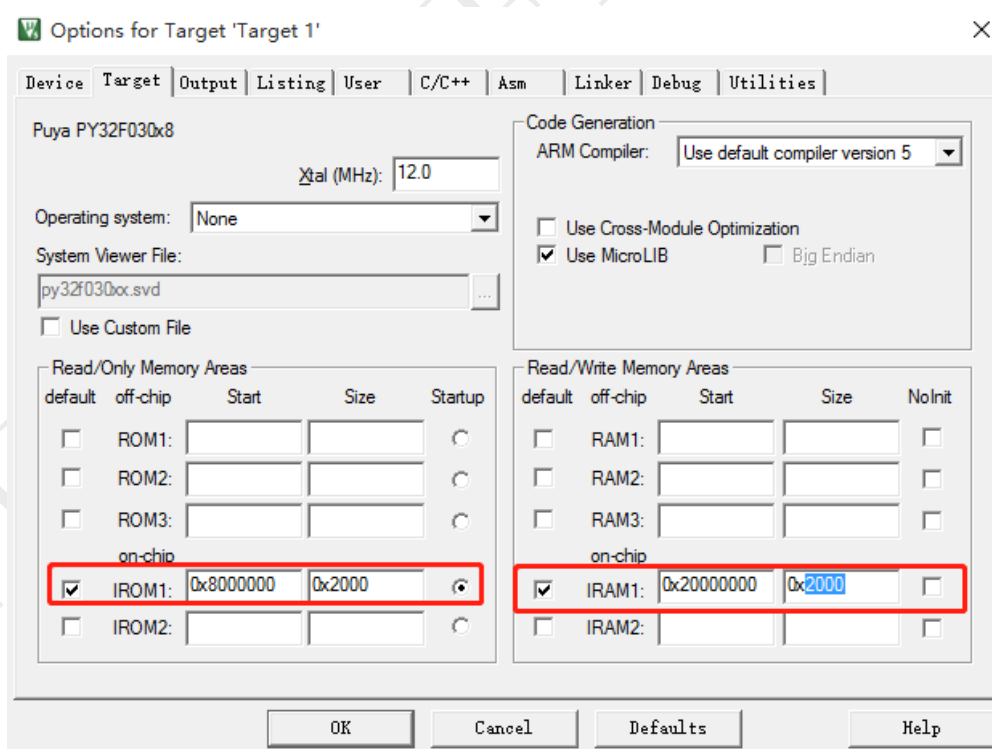
1.2.1 代码描述

```

30 int main(void)
31 {
32     BSP_PB_Init(BUTTON_KEY,BUTTON_MODE_GPIO);
33     /* Test if Key push-button on PY32F030 Start_kit Board is pressed */
34     if (BSP_PB_GetState(BUTTON_KEY) == GPIO_PIN_RESET)
35     {
36         /* If Key is pressed, execute user code */
37         // IAP_Init();
38     }
39     else
40     { /* Keep the user application running */
41
42         /* Test if user code is programmed starting from address "APPLICATION_ADDRESS" */
43         if (((__IO uint32_t*)APPLICATION_ADDRESS) & 0xFFE0000) == 0x20000000)
44         {
45             /* Jump to user application */
46             JumpAddress = *((__IO uint32_t*) (APPLICATION_ADDRESS + 4));
47             Jump_To_Application = (pFunction) JumpAddress;
48
49             /* Initialize user application's Stack Pointer */
50             __set_MSP((__IO uint32_t*) APPLICATION_ADDRESS);
51
52             /* Jump to application */
53             Jump_To_Application();
54         }
55     }
56
57     /* Infinite loop */
58     while (1)
59     {
60     }
61 }
62 }

```

1.2.2 MDK配置



1.3 APP代码

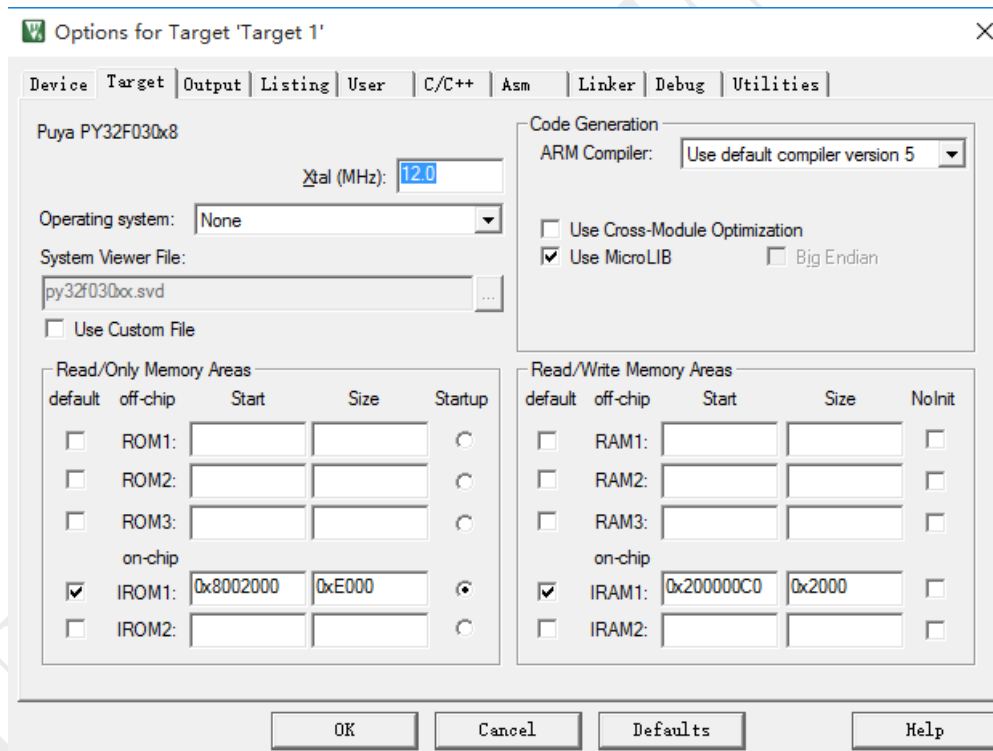
1.3.1 代码描述

```

31 int main(void)
32 {
33     uint32_t i;
34     //初始化所有外设, flash接口, systick
35     HAL_Init();
36
37     /* Copy the vector table from the Flash (mapped at the base of the application
38        load address 0x08002000) to the base address of the SRAM at 0x20000000. */
39     for(i = 0; i < 48; i++)
40     {
41         VectorTable[i] = *(__IO uint32_t*) (APPLICATION_ADDRESS + (i<<2));
42     }
43
44     SCB->VTOR = SRAM_BASE;
45     TIM14_init();
46
47     while (1)
48     {
49     }
50 }
51

```

1.3.2 MDK配置



2 版本历史

版本	日期	更新记录	作者
V1.0	2022.03.04	初版	LSQ



Puya Semiconductor Co., Ltd.

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